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LudoBench: Evaluating LLM Behavioural Decision-Making through Spot-Based Board Game Scenarios in Ludo

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Motivation

Objective: Evaluate whether LLMs can reason strategically under uncertainty by benchmarking their decisions on Ludo. Ludo is stochastic, multi-agent, multi-piece board game absent from prior benchmarks.

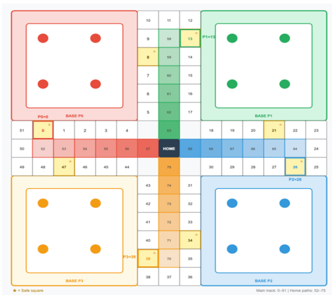


Fig 1: Annotated Ludo board

Why Ludo? Stochastic (dice), multi-agent (2–4), multi-piece (4 pieces), complete-information. It is a combination absent from chess, Go, poker, and Diplomacy.

Core Question: Can LLMs plan ahead, weigh competing objectives, and decide under dice-driven uncertainty?

Related work & Gap

- Prior LLM game work covers chess, Go, poker, Werewolf, Diplomacy. Stochastic race games are absent.
- PokerBench and GameBench provide legal moves to the model and report only aggregate win-rates.
- No prior benchmark jointly measures rule internalization, framing effects, and behavioural archetypes.

Key Findings

1) Rule compliance does not imply strategic competence. DeepSeek-Chat and Claude-3.5-Haiku both score <1% invalid, yet adopt opposite strategies. Rule compliance tells us nothing about how a model reasons about tradeoffs.
2) LLMs learn partial strategies, not complete ones. All six models agree with the GT agent only 40–46%. Each falls into a Finisher or Builder archetype, capturing only half of the balanced GT strategy.
3) Narrative framing shifts decisions on identical boards. Qwen-Plus changes 33% of its decisions under grudge framing on identical boards.

Methodology

- Spot-based evaluation: **480 handcrafted** single-decision boards across **12 categories** (capture, safe, home_entry, extra_turn, overshoot, blocked, 5 capture-tradeoff variants, grudge_paired), each isolating one strategic choice.
- Four agent types: Random (floor), Heuristic (one-step greedy with capture/safe bonuses), Game-Theory Optimal (depth-2 Expectiminimax / Expectimax-MaxN, ceiling), and the LLM .
- Prompt = board state + full rules + dice value; legal moves intentionally withheld so the model must internalize Ludo rules, making invalid-rate a measurable compliance signal.
- Paired grudge design presents identical boards with neutral vs. hostile-capture narratives, isolating history-conditioned framing from board-state effects.

Experimental Setup

- Six LLMs across four families: Qwen-Plus, Qwen-2.5-7B-Instruct, DeepSeek-Chat, Claude-3.5-Haiku, Llama-4-Scout, Gemma-3-12B-IT.
- 480 spots × 5 persona conditions (none, aggressive, greedy, safe, unforgiving) × 6 models = 14,400 evaluations; zero-shot.
- Behavioral metrics computed on valid outputs only; invalid rate recorded pre-fallback to keep rule compliance and decision quality as separate strata.
- GT baseline validated via head-to-head play (200 games per pairing): Random < Heuristic < GT (91% vs. Random, 59% vs. Heuristic).

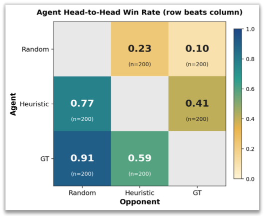


Figure 2: Agent skill ladder

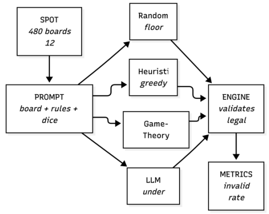


Figure 3: LudoBench evaluation pipeline

Results

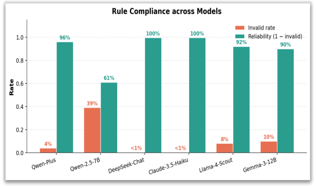


Fig 4: Rule compliance across six LLMs (invalid vs. reliability)

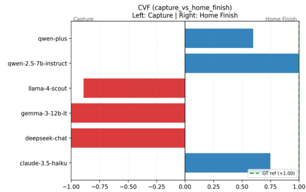
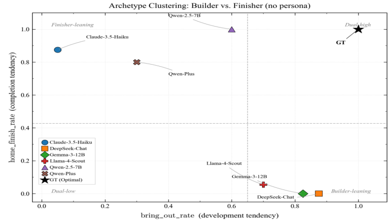


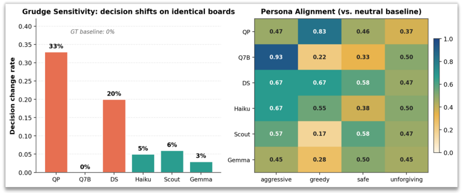
Fig 5: Tradeoff — Capture vs. Home Finish

DS & Haiku are <1% invalid. Q7B is 39% invalid; size alone does not predict compliance.



Finishers complete pieces well (Haiku 88%, Q7B 100%) but barely develop new ones. **Builders** develop aggressively (DS 88%, Gemma 82%) but never finish when capture is available. No LLM matches GT's balanced play.

Figure 6: Builder vs. Finisher archetypes



Qwen-Plus shifts decisions 33% on identical boards under grudge framing.
Persona steering is weak and unreliable.

Fig 7: Grudge sensitivity (left) and persona alignment heatmap (right)

Conclusion

- Rule compliance is not strategic skill. DS and Haiku both score <1% invalid yet adopt opposite play styles.
- LLMs learn half-strategies. Each is a Finisher or a Builder; only GT does both.
- Narrative framing shifts decisions by up to **33%** on identical boards.

Limitations & Future Work

- 40 spots per category gives point estimates; some ceiling effects.
- Single-decision evaluation only. No end-to-end multi-turn games tested.
- Future work: more models, non-English prompts, other stochastic race games.

Code & Dataset

Repository: anonymous.4open.science/r/LudoBench-5CBF/
Dataset: 480 engine-validated spots (40 per category) + 80 paired grudge scenarios. All code, spot data, and model outputs released.

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